

Program 3: Create a dockerized python flask or node.js application

File structure:

/Dockerfile

/package.json

/package-lock.json

/index.js

Create folder and initialize package.json

mkdir program4 > cd program4 > npm init -y > npm i express

Dockerfile

FROM node:18-slim

WORKDIR /app

COPY package*.json .

RUN npm install

COPY . .

EXPOSE 3000

CMD ["node", "index.js"]

index.js

```
const express = require('express')
```

```
const app = express()
```

```
app.get('/', (req, res) => res.send("Hello from 3rd program"))
```

```
app.listen(3000, () => { console.log("Server running on http://localhost:3000"); })
```

Run the following command to create docker image and container

sudo docker build -t program3 .

sudo docker run -d -p 3000:3000 --name pro3 program3

```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab3
[+] Building 49.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 155B
=> [internal] load metadata for docker.io/library/node:18-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd4225c1e51a564f87584b9bd21dd651838830e
=> => resolve docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd4225c1e51a564f87584b9bd21dd651838830e
=> sha256:101e0128c8ea9a6f6e5eb2abbae8486503c6383c35cb30e2c60842a5a288479 6.54kB / 6.54kB
=> sha256:61320b01ae5e0798393ef25f2d72cfaf43703e60ba089b07d7170acbabf8f62 28.23MB / 28.23MB
=> sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d2621b8273defb45455315 3.31kB / 3.31kB
=> sha256:b1831921e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee386c5c8527d 38.25MB / 38.25MB
=> sha256:f9ab18e354e6855ae56ef2b290dd4225c1e51a564f87584b9bd21dd651838830e 6.49kB / 6.49kB
=> sha256:fc3fa127a182135fd956e68d570b1932a758f8008866d8dd6e131cf89de9605 1.93kB / 1.93kB
=> sha256:c768ab8c8a73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021fc9a07a 1.71MB / 1.71MB
=> sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c3dc 0B / 448B
=> extracting sha256:61320b01ae5e0798393ef25f2d72cfaf43703e60ba089b07d7170acbabf8f62 1.2s
=> extracting sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d2621b8273defb45455315 0.0s
=> extracting sha256:b1831921e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee386c5c8527d 1.3s
=> extracting sha256:c768ab8c8a73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021fc9a07a 0.0s
=> extracting sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c3dc 0.0s
=> [internal] load build context
=> => transferring context: 43.58kB
=> [2/5] WORKDIR /app
=> [3/5] COPY package*.json .
=> [4/5] RUN npm install
=> [5/5] COPY . .
=> exporting to image
=> exporting layers
=> writing image sha256:4fccffca92f235584a198ba5042eae6253e2f81d34602f47fbdcc6c3b3a66b3b
=> naming to docker.io/library/program3
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab3$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
program4 latest 79b476c64819 2 seconds ago 195MB
program3 latest 4fccffca92f2 29 seconds ago 198MB
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab3$ sudo docker run -d -p 3000:3000 --name pro3 program3
e9a66b6cd9d47cd740431b6fd48a7db31f17b5df1f569ea3e9e3b8905baa39fc
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab3$ sudo docker container ls -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
e9a66b6cd9d4 program3 "docker-entrypoint.s..." 10 seconds ago Up 10 seconds 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp pro3
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab3$
```

Go to browser and type <http://localhost:3000> to see the output

To list and remove container and images

sudo docker container ls -a

```
sudo docker container stop <container id>  
sudo docker container rm<container id>  
sudo docker images  
sudo docker rmi <imageid>
```